

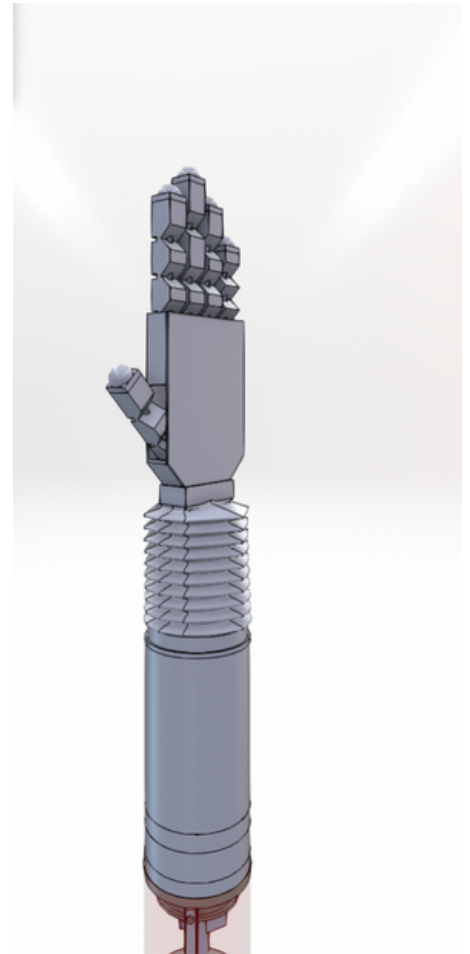
Soft robotic hand

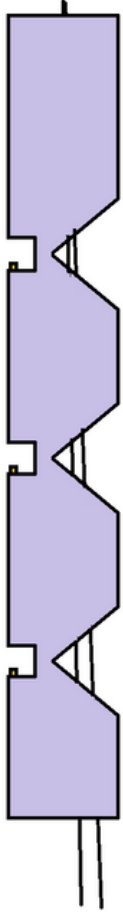
Guru Nishaanth S

Phase I : Design the hand part

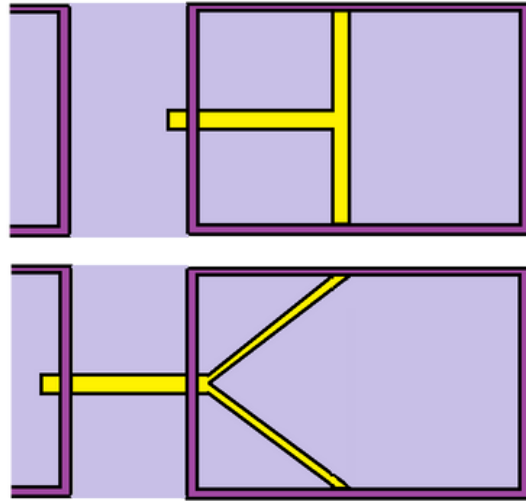
After reviewing some videos and other models, I would like to create a design with a single piece that has minimal moving parts (except for the tendons).

Based on earlier designs, I would like to make several design changes, including having two tendons per finger, incorporating a compliant mechanism between the palm and finger sections, and adding a few compliant mechanisms behind the finger to prevent it from fully bending.

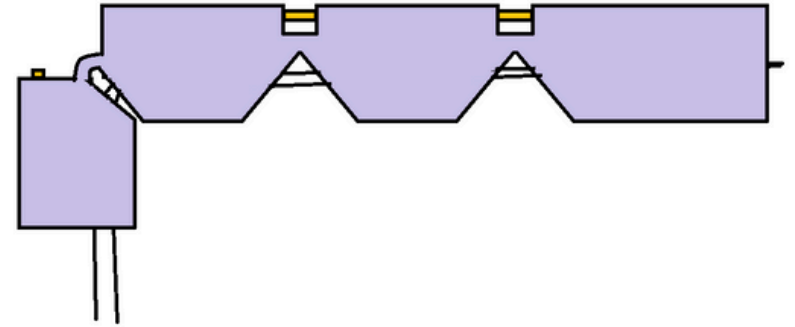




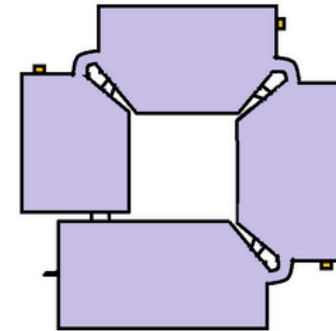
Single Finger



**Compliant
mechanism**

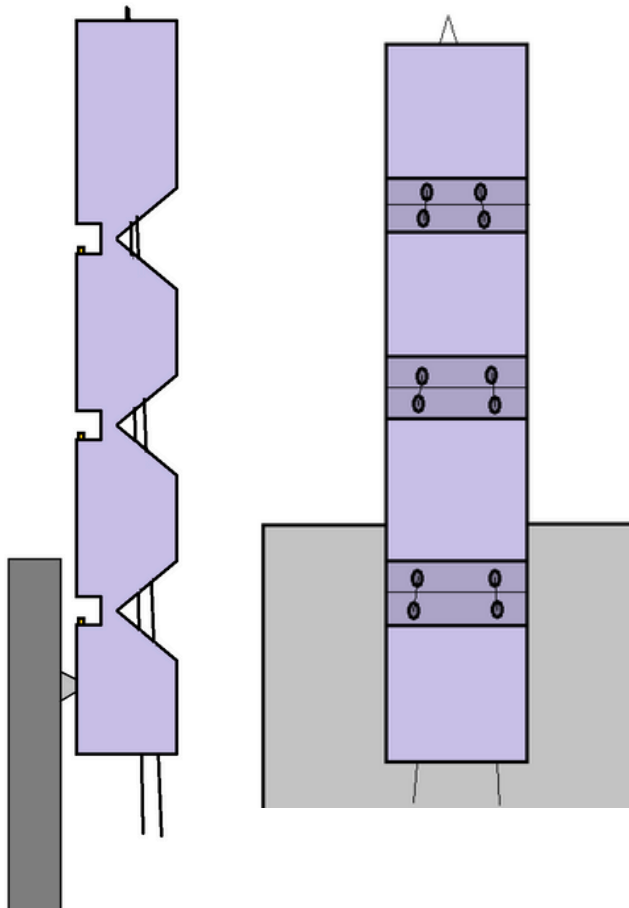


**Finger with the compliant
mechanism deployed**

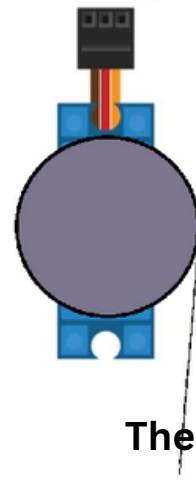


**Finger without the compliant
mechanism deployed**

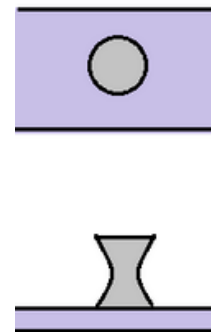
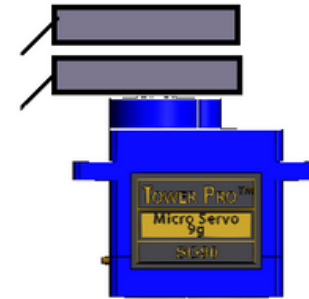
The finger must be printed in TPU, and the compliant mechanism must be in PLA



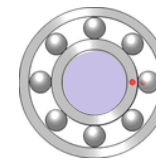
Finger attached to the palm



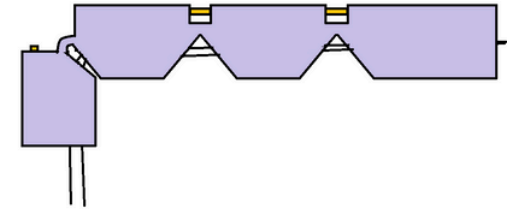
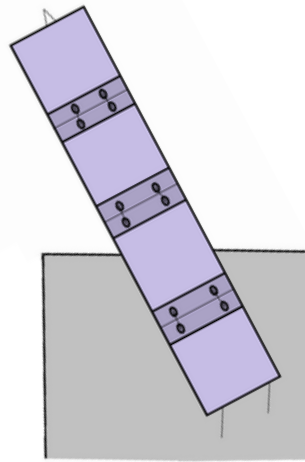
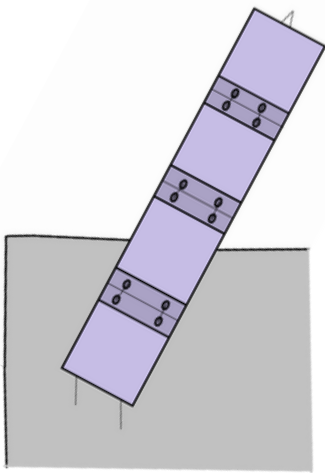
The gear mechanism is fixed with a Servo



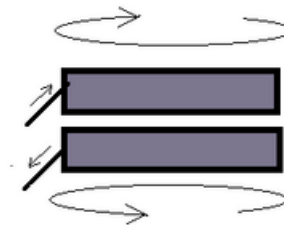
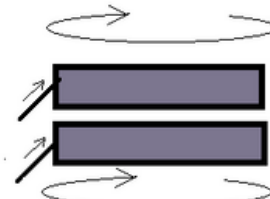
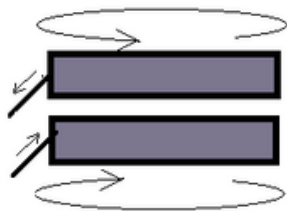
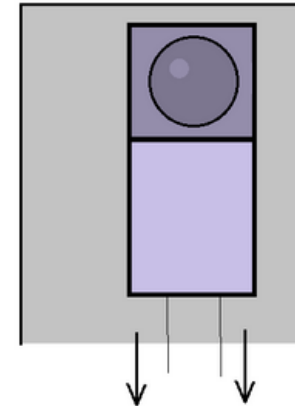
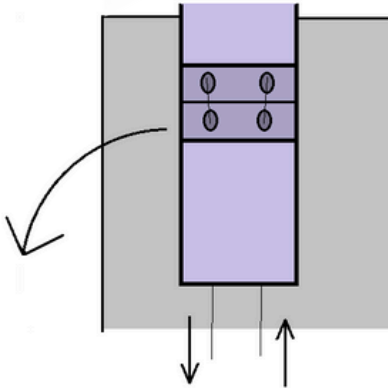
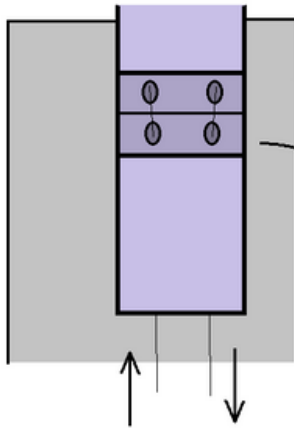
A compliant mechanism that is made of TPU to twist and move the finger sideways. this mechanism is to replace the ball bearing



I was thinking of using this gripper as nails for the hand

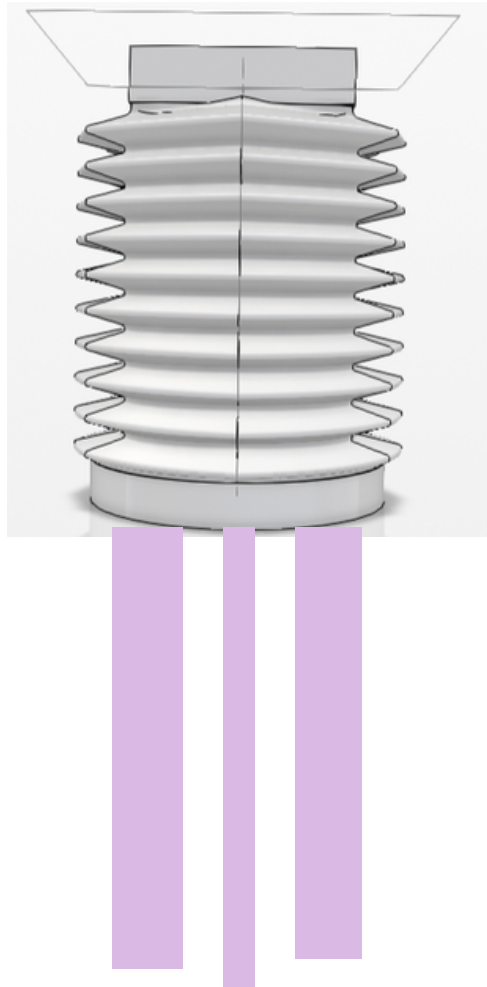


Finger when both strings are pulled

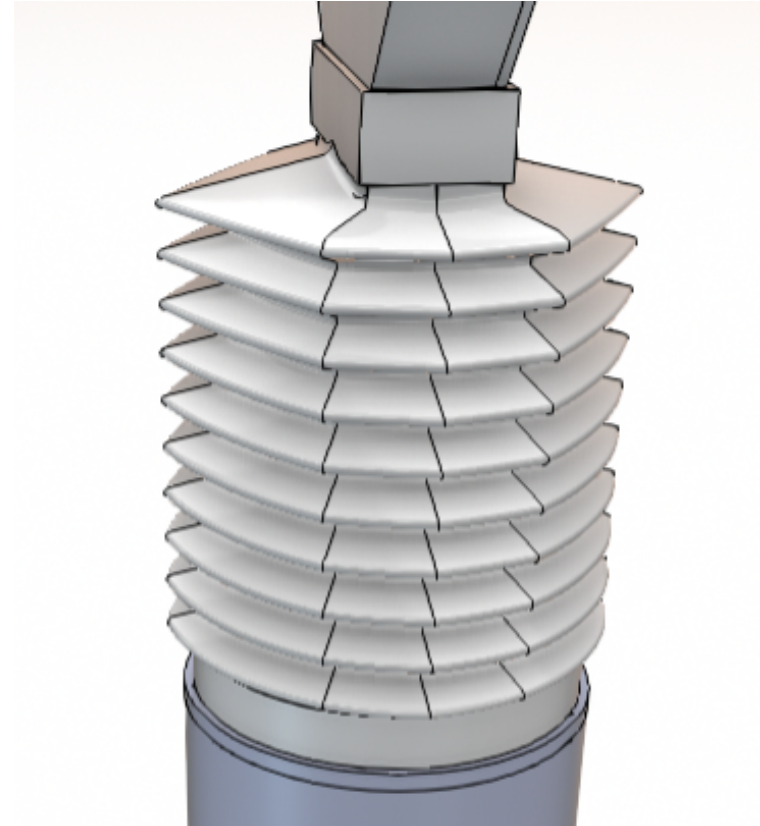


The movement of the gearbox and its subsequent motion of the finger

Soft robotic Wrist



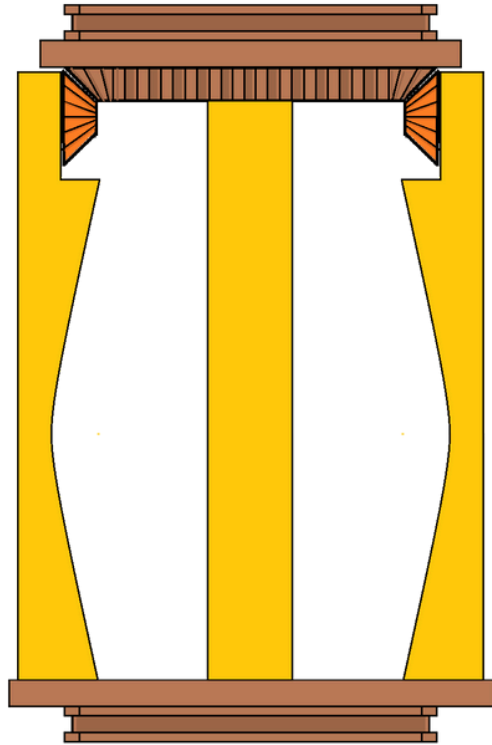
To have a wrist movement we need
3 actuators



A contraption that can connect
with the hand and the rest of the
arm

Soft robotic Arm (if time permits)

To redesign a lot of parts and mechanisms of this basic design of the soft robot. Mainly where I used a ball bearing. I would like to use a gear mechanism like the one shown below. And also have 4 of these for stability



Phase II Print and assemble the parts

- The main path of this phase is to decide the materials and prototype the designs if they work with the motor.
- First, print a single finger to see if the design actually works.
- If it works, then print the whole hand design and see if it comes together.
- Then fix the motors and complete the hand.
- Continue the similar process for the wrist and arm design.

Phase III Simple humanoid hand movements

- To make some simple movements, like the index finger and the thumb. Pincer grip
- Abduction of all fingers(test of the finger compliant mechanism)
- Adduction of all fingers (the test for the palm to finger mechanism)
- Flexion of all fingers(normal movement of fingers)
- Opposition for all fingers(full test for the hand)

Phase IV Inverse and Forward kinematics

- To make a proper movement of the whole, I would like to do an inverse and forward kinematics of the whole hand.
- This can help to understand all the motions that can be done by the hand.

Phase V Computer vision

- In this method, we can use a camera to take the hand gesture as an input and then reproduce the gesture to the robotic hand
- This can help to make different movements of the hand, and this can be trained and recreated at will.



References

- <https://www.youtube.com/watch?v=sOVMLvNhsJ8>
- <https://www.dexhand.org/>
- [Handbook of Compliant Mechanisms - download](#)
- <https://www.researchgate.net/publication/224199787>
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- [https://www.researchgate.net/publication/230739692 Virtual human hand model and kinematics](https://www.researchgate.net/publication/230739692_Virtual_human_hand_model_and_kinematics)
- https://elib.dlr.de/100591/1/Stillfried_phd_main_v1_4.pdf